

HDOC – Experiment 38

Question

Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

❖ Algorithm

1 Start the program.

2 Declare a character array to store the string.

3 Accept a string from the user using `fgets()`.

4 Display the menu:

- Count spaces, digits, and special characters.
- Count frequency of a given character.
- Toggle the case of each character.
- Exit.

5 Read the user's choice.

6 Use a switch statement:

- **Choice 1:** Call `countCharacters()` to count spaces, digits, and special characters.
- **Choice 2:** Ask for a character and call `frequency()` to count its occurrences.
- **Choice 3:** Call `toggleCase()` to change uppercase letters to lowercase and lowercase letters to uppercase.

- **Choice 4:** Exit the program.
- Otherwise, display an invalid-choice message.

7 Repeat the menu until the user selects Exit.

8 Stop.

❖ Program code in C

```
#include <stdio.h>
```

```
void countCharacters(char str[]);
```

```
void frequency(char str[]);
```

```
void toggleCase(char str[]);
```

```
int main()
```

```
{
```

```
    char str[200];
```

```
    int choice;
```

```
    printf("Enter a string: ");
```

```
    fgets(str, sizeof(str), stdin);
```

```
    do
```

```
    {
```

```
        printf("\n----- MENU ----- \n");
```

```
        printf("1. Count spaces, digits and special characters\n");
```

```
        printf("2. Count frequency of a given character\n");
```

```
        printf("3. Toggle the case of each character\n");
```

```
printf("4. Exit\n");
```

```
printf("Enter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch (choice)
```

```
{
```

```
    case 1:
```

```
        countCharacters(str);
```

```
        break;
```

```
    case 2:
```

```
        frequency(str);
```

```
        break;
```

```
    case 3:
```

```
        toggleCase(str);
```

```
        break;
```

```
    case 4:
```

```
        printf("Exiting program...\n");
```

```
        break;
```

```
    default:
```

```
        printf("Invalid choice! Please enter a choice from 1 to 4.\n");
```

```
}
```

```
} while (choice != 4);
```

```
    return 0;
}
```

```
/* Function to count spaces, digits and special characters */
```

```
void countCharacters(char str[])
```

```
{
```

```
    int i = 0;
```

```
    int spaces = 0;
```

```
    int digits = 0;
```

```
    int special = 0;
```

```
    while (str[i] != '\0')
```

```
    {
```

```
        if (str[i] == ' ')
```

```
        {
```

```
            spaces++;
```

```
        }
```

```
        else if (str[i] >= '0' && str[i] <= '9')
```

```
        {
```

```
            digits++;
```

```
        }
```

```
        else if (!((str[i] >= 'A' && str[i] <= 'Z') ||
```

```
                    (str[i] >= 'a' && str[i] <= 'z') ||
```

```
                    str[i] == '\n'))
```

```
        {
```

```
            special++;
```

```

    }

    i++;
}

printf("\nNumber of spaces = %d\n", spaces);
printf("Number of digits = %d\n", digits);
printf("Number of special characters = %d\n", special);
}

/* Function to find frequency of a given character */
void frequency(char str[])
{
    char ch;
    int i = 0;
    int count = 0;

    printf("Enter the character to search: ");
    scanf(" %c", &ch);

    while (str[i] != '\0')
    {
        if (str[i] == ch)
        {
            count++;
        }
    }
}

```

```

        i++;
    }

    printf("Frequency of '%c' = %d\n", ch, count);
}

/* Function to toggle the case of each alphabet */
void toggleCase(char str[])
{
    int i = 0;

    printf("String after toggling case: ");

    while (str[i] != '\0')
    {
        if (str[i] >= 'A' && str[i] <= 'Z')
        {
            printf("%c", str[i] + 32);
        }
        else if (str[i] >= 'a' && str[i] <= 'z')
        {
            printf("%c", str[i] - 32);
        }
        else
        {
            printf("%c", str[i]);
        }
    }
}

```

```

    i++;
}
}

```

❖ Test case table

Test Case	Input String	Choice/Input	Expected Result
1	Hello 123!	Choice 1	Spaces = 1, Digits = 3, Special characters = 1
2	banana	Choice 2, Character = a	Frequency of a = 3
3	Hello World123!	Choice 3	hELLO wORLD123!
4	ABC xyz	Choice 3	abc XYZ
5	C@SE 2026#	Choice 1	Spaces = 1, Digits = 4, Special characters = 2
6	Programming	Choice 2, Character = g	Frequency of g = 2
7	Any string	Choice 4	Program exits
8	Any string	Choice 7	Invalid choice!

❖ Compile, Execute and Evaluate

```
(kali㉿kali)-[~/Desktop/exp38]
$ vi string3.c

(kali㉿kali)-[~/Desktop/exp38]
$ gcc string3.c -o string3

(kali㉿kali)-[~/Desktop/exp38]
$ ./string3
Enter a string: Hello 123!

—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 1

Number of spaces = 1
Number of digits = 3
Number of special characters = 1

—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 4
Exiting program...

(kali㉿kali)-[~/Desktop/exp38]
$ ./string3
Enter a string: banana

—— MENU ——
1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 2
```

```
Enter the character to search: a
Frequency of 'a' = 3
```

```
—— MENU ——
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 4
```

```
Exiting program...
```

```
(kali㉿kali)-[~/Desktop/exp38]
```

```
$ ./string3
```

```
Enter a string: Hello World123!
```

```
—— MENU ——
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 3
```

```
String after toggling case: hELLO wORLD123!
```

```
—— MENU ——
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 4
```

```
Exiting program...
```

```
(kali㉿kali)-[~/Desktop/exp38]
```

```
$ ./string3
```

```
Enter a string: ABC xyz
```

```
—— MENU ——
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character

```
4. Exit
Enter your choice: 3
String after toggling case: abc XYZ
```

```
—— MENU ——
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 4
Exiting program...
```

```
(kali㉿kali)-[~/Desktop/exp38]
```

```
$ C@SE 2026#
```

```
C@SE: command not found
```

```
(kali㉿kali)-[~/Desktop/exp38]
```

```
$ ./string3
```

```
Enter a string: C@SE 2025#
```

```
—— MENU ——
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 1
```

```
Number of spaces = 1
```

```
Number of digits = 4
```

```
Number of special characters = 2
```

```
—— MENU ——
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 4
```

```
Exiting program...
```

```
(kali㉿kali)-[~/Desktop/exp38]
```

```
$ ./string3
```

```
Enter a string: Programming
```

```
——— MENU ———
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 2
```

```
Enter the character to search: g
```

```
Frequency of 'g' = 2
```

```
——— MENU ———
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 4
```

```
Exiting program...
```

```
(kali㉿kali)-[~/Desktop/exp38]
```

```
$ ./string3
```

```
Enter a string: good
```

```
——— MENU ———
```

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

```
Enter your choice: 4
```

```
Exiting program...
```

```
(kali㉿kali)-[~/Desktop/exp38]
```

```
$ ./string3
```

```
Enter a string: making
```

```
——— MENU ———
```

1. Count spaces, digits and special characters

—— MENU ——

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

Enter your choice: 7

Invalid choice! Please enter a choice from 1 to 4.

—— MENU ——

1. Count spaces, digits and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit

Enter your choice: 4

Exiting program...

(kali@kali)-[~/Desktop/exp38]
\$